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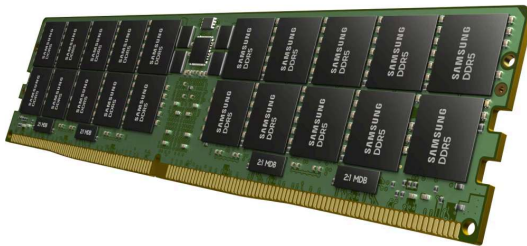
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Tech Blog

Innovative Memory Solution: Samsung's MCRDIMM Targets High-Performance Computing



Jun 16, 2024

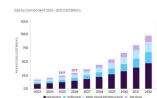


Samsung has developed a Multi-Ranked Buffered Dual In-Line Memory Module (MCRDIMM), which provides more memory and bandwidth without increasing memory slots on the server board. This module doubles the bandwidth of existing DRAM components by combining two [DDR5](#) components to deliver data transmission speeds of up to 8.8 Gb/s. MCRDIMM is expected to be actively adopted for AI applications that require High-Performance Computing (HPC) to process data and perform complex calculations at high speeds.

MCRDIMM Features and Benefits

MCRDIMM enables CPU to access two ranks of memory simultaneously on a single DIMM with a chip-set layout of two identical DDR5 DIMMS. The benefit of this approach is that the DRAM devices are not expected to operate at a faster clock frequency. By accessing two DRAM devices at the same time, the CPU can effectively double the memory bandwidth coming from the DIMM. Accessing the two DRAM devices is accomplished by using a special mux data buffer resident on

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the DRAM, which allows for simultaneous access to both DIMMs thereby doubling the data rate to the host.

Summary

Samsung’s first-generation MCRDIMMs offer data transfer rates of up to 8.8 Gb/s. Samsung is currently sampling 16Gb Mono MCRDIMM devices with enhanced performance, capacity, and power consumption. With this [innovative memory solution](#), Samsung is advancing the next frontier for Artificial Intelligence (AI), Machine Learning (ML), and Large Language Model (LLM) processing.

#MCRDIMM #DDR5 #HPC #ML #LLM #MemoryModule #Memory #Module
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